A Process-Relational Presentation of Finite Metric Geometry

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Abstract

This paper develops a conceptual framework for finite discrete geometry organized around three commitments: geometric structure arises from generators and composition laws rather than from a background space; objects are equivalence classes of finite processes under a congruence; and metric structure is an external parameter rather than an intrinsic property of geometric objects.

The mathematical language used throughout is that of quotients of free categories on weighted directed graphs — structures well-established in geometric group theory, the theory of groupoids, and enriched category theory. The principal constructions (metric emergence from path optimization, universality for finite metric spaces, categorical equivalence with Lawvere metric spaces, Gromov–Hausdorff compatibility) are instances of known results, recast within this framework and collected in one place. The relationship to existing work is stated explicitly in Section 11.

The main contribution of the paper is a *presentation-level semantic framework*: geometric structure is defined at the level of generators and congruence laws, prior to quotienting, and metric properties arise as invariants of congruence classes of processes. This separates layers — syntax (generators), laws (congruence), objects (quotient), and distance (emergent metric) — that are usually conflated in classical geometry and in geometric group theory expositions.

The paper presents a complete isomorphism classification of single-generator congruences by the pair (n, k) (Proposition 4.10). For *non-invertible* systems this classification is non-trivial: the quotients are cyclic monoids $M(n, k)$ and the pair (n, k) is a genuine complete invariant. This is an instance of a classical result in monoid theory [3] recast in relational-process language. For *invertible* systems the constraint $k = 0$ is forced (Corollary 4.12), the pseudo-periodic case does not arise, and the result reduces to the elementary classification of cyclic groups by order.

The separation of layers makes explicitly askable a question that classical treatments obscure: what geometric structure does a specific congruence law produce in the nerve of the free category, independently of the quotient it defines?

This question is developed in Section 12, the principal new contribution of this paper. Given a group presentation $\langle R \mid \mathcal{L} \rangle$, we define a depth filtration on both the nerve of the free group $F(R)$ and the nerve of the quotient group $G = F(R)/\!\sim$. Treating this filtration as a persistence module in the sense of topological data analysis, we prove three structural theorems: the free group filtration has exactly $|R|$ homology classes in degree 1, each born at depth 1 with infinite persistence, and nothing in higher degrees; a commutativity law $a \cdot b \sim b \cdot a$ creates a new class in H_2 at depth 2 with infinite persistence; and a torsion law $a^n \sim \mathrm{id}$ modifies the persistence of the generator's H_1 class at depth n. The *presentation persistence diagram*, defined as the difference between the free and quotient persistence diagrams, is an invariant of the presentation that records when and how each congruence law reshapes the topology of the classifying space. An open-source implementation is described that computes this comparison directly from a presentation, making the invariant accessible without triangulation.

Scope and Organization

A reader familiar with geometric group theory or enriched category theory will recognize the structures below as weighted directed graphs, path metrics, and their categorical formulation via Lawvere's identification of metric spaces with enriched categories [1]. The exposition is self-contained: all definitions are given from scratch, and no prior knowledge of category theory is assumed. The framework's relationship to existing work — including what is reformulated and what is genuinely new — is stated in Section 11.

Notation. V denotes a set of states, R a set of generators, $s, t : R \to V$ source and target maps, $\mathcal{C}$ the set of all configurations over (V, R, s, t), and $\sim$ a congruence on $\mathcal{C}$. We write $[C]$ for the equivalence class of a configuration C, and $\mathcal{C}/{\sim}$ for the quotient. The empty configuration at state v is id_v, written id when v is clear from context.

1 Relational Systems

Definition 1.1 (Relational System)**.** A *relational system* is a triple (V, R, s, t) where:

- V is a non-empty set of *states*,
- R is a non-empty set of *generators*,
- $s, t : R \to V$ are the *source* and *target* maps, assigning to each generator a domain and codomain state.

Definition 1.2 (Compatibility)**.** Generators $r_1, r_2 \in R$ are *compatible* if $t(r_1) = s(r_2)$.

Remark 1.3 (Graph structure)*.* A relational system (V, R, s, t) is precisely a directed graph with vertex set V and edge set R. Generators are edges; compatibility is the standard condition for path concatenation; configurations are directed paths. A weight function $w : R \to [0, \infty)$ may be added when metric structure is needed (Section 5). In Section 8 this graph is the generating data for a free enriched category, and the congruence $\sim$ plays the role of relations in the categorical sense of a *presentation by generators and relations*.

2 Configurations as Finite Processes

Definition 2.1 (Configuration)**.** A *configuration* is a finite sequence of generators

$$C = (r_1, r_2, \dots, r_n), \quad r_i \in R,$$

such that $t(r_i) = s(r_{i+1})$ for each $i = 1, \dots, n-1$ (each consecutive pair is compatible in the sense of Definition 1.2). The *source* and *target* of C are $s(C) := s(r_1)$ and $t(C) := t(r_n)$. The *empty configuration* id_v at state v (of length 0) satisfies $s(\mathrm{id}_v) = t(\mathrm{id}_v) = v$ and acts as a two-sided identity. We write id when the state is clear from context. The set of all configurations is denoted $\mathcal{C}$.

Definition 2.2 (Configuration Composition)**.** For $C = (r_1, \dots, r_m)$ and $D = (s_1, \dots, s_n)$ with $t(C) = s(D)$ (i.e., $t(r_m) = s(s_1)$), their *composition* is concatenation:

$$C \circ D = (r_1, \dots, r_m, s_1, \dots, s_n).$$

Composition is associative and $\mathrm{id}_v \circ C = C$, $C \circ \mathrm{id}_v = C$ whenever the sources and targets match.

Remark 2.3 (Configurations are paths)*.* A configuration is precisely a directed path in the graph (V, R, s, t) of Definition 1.1. Composition is path concatenation, which is associative by construction. No further associativity axiom is required.

3 Congruence and Observational Indistinguishability

Definition 3.1 (Congruence)**.** An equivalence relation $\sim$ on $\mathcal{C}$ is a *congruence* (with respect to composition) if it satisfies:

(i) **Typing preservation.** $C_1 \sim C_2$ implies $s(C_1) = s(C_2)$ and $t(C_1) = t(C_2)$. Congruent configurations share the same source and target state.

(ii) **Composition compatibility.** For all $C_1, C_1', C_2, C_2' \in \mathcal{C}$,

$$C_1 \sim C_1' \text{ and } C_2 \sim C_2' \;\Rightarrow\; C_1 \circ C_2 \sim C_1' \circ C_2',$$

whenever the compositions are defined.

Typing preservation ensures that the quotient composition $[C_1]\circ[C_2]$ is well-typed: compatible classes remain compatible after quotienting.

Remark 3.2 (Practical scope of congruences)*.* The definition above allows arbitrary congruences. In practice, the paper restricts attention to congruences *generated by a finite set of relational laws* — that is, the smallest congruence containing a given set of pairs (C_i, C_i'), closed under the conditions above. This is the setting of category presentations. Additional properties — such as confluence (the Church–Rosser property) or termination of the associated rewriting system — may be imposed when decidability or algorithmic metric computation is required. Without such properties, membership in a congruence class may be undecidable and metric values may be uncomputable. The examples throughout the paper use finitely generated congruences with straightforward normal forms.

Definition 3.3 (Observational Indistinguishability)**.** Given a congruence $\sim$ on $\mathcal{C}$, configurations C_1 and C_2 are *observationally indistinguishable with respect to* $\sim$ if, for every $D \in \mathcal{C}$:

$$C_1 \circ D \sim C_2 \circ D \quad \text{(whenever } C_1, C_2 \text{ are left-compatible with } D\text{)},$$

$$D \circ C_1 \sim D \circ C_2 \quad \text{(whenever } D \text{ is left-compatible with } C_1, C_2\text{)}.$$

Remark 3.4 (Derived property)*.* Observational indistinguishability is a consequence of congruence, not a definition of it. If $C_1 \sim C_2$ and $\sim$ is a congruence, then for any compatible D: $C_1 \circ D \sim C_2 \circ D$ follows immediately from Definition 3.1 applied with $D \sim D$. The intended workflow is: (1) fix a relational system; (2) specify $\sim$ by listing relational laws; (3) verify Definition 3.1; (4) conclude that equivalent configurations are observationally indistinguishable.

Definition 3.5 (Relational Inverse)**.** Given a relational system (V, R, s, t) with congruence $\sim$, a generator $r \in R$ has an *inverse* $r^{-1} \in R$ if:

$$(r, r^{-1}) \sim \mathrm{id}_{t(r)} \quad \text{and} \quad (r^{-1}, r) \sim \mathrm{id}_{s(r)}.$$

A relational system is *invertible* (with respect to $\sim$) if every generator has an inverse. Note that r^{-1} must satisfy $s(r^{-1}) = t(r)$ and $t(r^{-1}) = s(r)$ for the configurations (r, r^{-1}) and (r^{-1}, r) to be well-formed.

4 Objects and the Quotient Construction

Definition 4.1 (Object)**.** An *object* is an equivalence class $[C]$ of a configuration C under a congruence $\sim$:

$$[C] = \{C' \in \mathcal{C} \mid C' \sim C\}.$$

The set of all objects is the quotient $\mathcal{C}/{\sim}$.

Remark 4.2 (Terminology and categorical context)*.* In the categorical language of Section 8, the equivalence classes $[C]$ are *morphisms* in the quotient category, while the states V of the relational system serve as categorical *objects*. The term "object" is used here in the geometric sense—an equivalence class of processes that no relational test can distinguish—not in the categorical sense. The two usages are reconciled in Section 8.

Theorem 4.3 (Quotient Construction)**.** *Let (V, R, s, t) be a relational system with configuration set $\mathcal{C}$, and let $\sim$ be a congruence on $\mathcal{C}$. Then:*

(i) The quotient $\mathcal{C}/{\sim}$ is well-defined as a set of equivalence classes.

(ii) Composition descends to a well-defined partial operation on $\mathcal{C}/{\sim}$: whenever $t(C_1) = s(C_2)$,

$$[C_1] \circ [C_2] \;:=\; [C_1 \circ C_2].$$

(iii) The induced composition is associative.

(iv) For each state $v \in V$, the class $[\mathrm{id}_v]$ acts as a two-sided identity for all classes $[C]$ with $s(C) = v$ or $t(C) = v$.

Proof. **(i).** $\sim$ partitions $\mathcal{C}$ into disjoint non-empty equivalence classes.

(ii). Suppose $[C_1] = [C_1']$ and $[C_2] = [C_2']$, i.e., $C_1 \sim C_1'$ and $C_2 \sim C_2'$. Since $\sim$ is a congruence, $C_1 \circ C_2 \sim C_1' \circ C_2'$, so $[C_1 \circ C_2] = [C_1' \circ C_2']$. The operation is independent of representatives. Compatibility of $[C_1]$ and $[C_2]$ is inherited from the compatibility of their representatives.

(iii). For $C_1, C_2, C_3 \in \mathcal{C}$, associativity at the configuration level gives $(C_1 \circ C_2) \circ C_3 = C_1 \circ (C_2 \circ C_3)$. By part (ii):

$$([C_1] \circ [C_2]) \circ [C_3] = [(C_1 \circ C_2) \circ C_3] = [C_1 \circ (C_2 \circ C_3)] = [C_1] \circ ([C_2] \circ [C_3]).$$

(iv). For any C starting at v, $\mathrm{id}_v \circ C = C$, hence $[\mathrm{id}_v] \circ [C] = [C]$. Similarly on the right. □

4.1 Taxonomy of Single-Generator Quotients

Proposition 4.4 (Conditions for a cyclic closed quotient)**.** *Let (V, R, s, t) be a relational system with a single generator a and congruence $\sim$. The quotient $\mathcal{C}/{\sim}$ is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ if and only if:*

(i) ***Closure.*** $a^n \sim \mathrm{id}$ *for some* $n \geq 1$.

(ii) ***Uniqueness.*** $a^j \not\sim a^k$ *for all* $0 \leq j < k < n$.

When closure holds but uniqueness fails, the quotient has period strictly less than n. When closure fails, all powers are distinct.

Proof. If both conditions hold, the n classes $[a^0], \ldots, [a^{n-1}]$ are distinct and $\mathcal{C}/{\sim} \cong \mathbb{Z}/n\mathbb{Z}$ by the congruence property. If $a^j \sim a^k$ for some $j < k < n$, the quotient has period $k - j < n$. If no n satisfies $a^n \sim \mathrm{id}$, all powers are distinct. □

Lemma 4.5 (Reduction to minimal period)**.** *Let (V, R, s, t) be an* invertible *relational system (with respect to a congruence $\sim$) with single generator a. If $a^n \sim \mathrm{id}$ for some $n \geq 1$ but the configurations $a^0, a^1, \ldots, a^{n-1}$ are not all pairwise inequivalent, then there exists a strictly smaller $k < n$ such that $a^k \sim \mathrm{id}$.*

Proof. If intermediate configurations are not all distinct, there exist $i < j \leq n$ with $a^i \sim a^j$. By the congruence property, composing both sides on the right with a^{-i} gives $a^i \circ a^{-i} \sim a^j \circ a^{-i}$, i.e., $\mathrm{id} \sim a^{j-i}$. Set $k := j - i < n$. Repeating if necessary yields a minimal period with all intermediate configurations pairwise inequivalent. □

Proposition 4.6 (Taxonomy of closure)**.** *Let (V, R, s, t) be a relational system with single generator a and congruence $\sim$, at its minimal period (all intermediate configurations distinct). Exactly one of the following holds:*

(i) ***Closed.*** $a^n \sim \mathrm{id}$ *for some $n \geq 1$. Quotient: $\mathbb{Z}/n\mathbb{Z}$.*

(ii) ***Eventually periodic.*** $a^n \sim a^k$ *for some $n > k \geq 1$ (so the orbit revisits $[a^k]$ without returning to* id*). Quotient: the cyclic monoid $M(n, k)$.*

(iii) ***Open.*** *No periodicity law holds; all powers $a^0, a^1, \ldots$ are distinct. Quotient: $\mathbb{N}$ (monoid) or $\mathbb{Z}$ (group, if invertible).*

Proof. By Lemma 4.5 (applied without the invertibility assumption in cases (i) and (ii)) it suffices to work at minimal n. Cases (i) and (ii) are distinguished by whether the return target is $[a^0] = [\mathrm{id}]$ or some $[a^k]$ with $k \geq 1$. They cannot both hold simultaneously. Case (iii) is the negation of (i) and (ii). The three cases are mutually exclusive and exhaustive. □

Remark 4.7 (Invertible systems and the vanishing of case (ii)). For *invertible* relational systems, case (ii) cannot occur. If $a^n \sim a^k$ with $k \geq 1$, composing on the right with a^{-k} (which exists by invertibility) gives $a^{n-k} \sim \mathrm{id}$. Since $n - k < n$, this contradicts the assumption that n is minimal. Therefore, for invertible systems, the taxonomy reduces to two cases: closed ($k = 0$, quotient $\mathbb{Z}/n\mathbb{Z}$) or open (quotient $\mathbb{Z}$). The pseudo-periodic case is a phenomenon of non-invertible systems only.

4.2 Classification of Single-Generator Congruences

The correct setting for a non-trivial classification by (n, k) is the *general* (possibly non-invertible) single-generator system. We develop this case first and then state the invertible result as a corollary.

Definition 4.8 (Type of a single-generator congruence)**.** Let (V, R, s, t) be a relational system with single generator a (not necessarily invertible) and congruence $\sim$.

(i) The congruence is *open* if all powers $a^0, a^1, a^2, \ldots$ are pairwise inequivalent.

(ii) The congruence is *eventually periodic* if there exist integers $n > k \geq 0$, both minimal subject to $a^n \sim a^k$. The pair (n, k) is the *type* of the congruence; $p := n - k \geq 1$ is the *period* and k is the *index*.

Remark 4.9 (Cyclic monoid structure)*.* When the congruence has type (n, k), the quotient $\mathcal{C}/\!\sim$ has exactly n elements $\{[a^0], [a^1], \ldots, [a^{n-1}]\}$, with multiplication determined by $[a^n] = [a^k]$. This is the *cyclic monoid* $M(n, k)$. Its structure has two parts: a "tail" $\{[a^0], \ldots, [a^{k-1}]\}$ of length k, and a "kernel cycle" $\{[a^k], \ldots, [a^{n-1}]\}$ of length $p = n - k$. The kernel cycle is a cyclic group of order p with identity element $[a^n] = [a^k]$. When $k = 0$, the tail is empty and $M(n, 0) \cong \mathbb{Z}/n\mathbb{Z}$ is a cyclic group.

Proposition 4.10 (Complete congruence classification)**.** *Let $\sim_1$ and $\sim_2$ be eventually-periodic congruences on single-generator systems with types (n_1, k_1) and (n_2, k_2). The quotients $\mathcal{C}/\!\sim_1$ and $\mathcal{C}/\!\sim_2$ are isomorphic as monoids if and only if $(n_1, k_1) = (n_2, k_2)$.*

Proof. *($\Leftarrow$)* If $(n_1, k_1) = (n_2, k_2)$, define $\Phi : \mathcal{C}/\!\sim_1 \to \mathcal{C}/\!\sim_2$ by $\Phi([a^j]_1) := [a^j]_2$. *Well-definedness*: if $a^j \sim_1 a^{j'}$ then j and j' reduce to the same residue modulo the congruence of type (n_1, k_1), which is the same as the congruence of type (n_2, k_2), so $a^j \sim_2 a^{j'}$. *Bijectivity*: the inverse is $\Psi([a^j]_2) := [a^j]_1$ by the same argument. *Composition-preservation*: $\Phi([a^j] \circ [a^m]) = [a^{j+m}]_2 = [a^j]_2 \circ [a^m]_2$.

($\Rightarrow$) Let $\Phi : \mathcal{C}/\!\sim_1 \to \mathcal{C}/\!\sim_2$ be an isomorphism. Since Φ preserves the monoid identity $[a^0] \mapsto [a^0]$ and multiplication, the element $\Phi([a]) = [a]_2$ (the only generator of the cyclic quotient). The isomorphism therefore maps $[a^j]_1 \mapsto [a^j]_2$ for all j. The total number of distinct elements in $\mathcal{C}/\!\sim_i$ is n_i, so bijectivity forces $n_1 = n_2$. The index k_i is the smallest j such that $[a^j]$ lies in the kernel cycle (the maximal subgroup); since Φ is an isomorphism, it maps the kernel cycle of $\mathcal{C}/\!\sim_1$ bijectively to that of $\mathcal{C}/\!\sim_2$, forcing $k_1 = k_2$. □

Reference: This result is an instance of the classical classification of cyclic monoids; see Clifford and Preston [3], Vol. 1, §1.2. □

Remark 4.11 (The (n, k) invariant)*.* The type (n, k) is a complete isomorphism invariant for eventually-periodic single-generator congruences: it determines the quotient up to isomorphism and is determined by it. The parameter n is the total number of equivalence classes. The parameter k is the length of the non-group tail: when $k = 0$ the quotient is a cyclic group; as k increases, more elements lie outside the group kernel, encoding longer transient behaviour before periodicity sets in. For multi-generator systems, no complete invariant analogous to (n, k) is known, and the corresponding word problem for finitely presented monoids is in general undecidable [7].

Corollary 4.12 (Classification for invertible systems)**.** *For invertible single-generator relational systems ($R = \{a, a^{-1}\}$), the index is always $k = 0$. The quotient is either $\mathbb{Z}$ (open case) or $\mathbb{Z}/n\mathbb{Z}$ for the minimal n with $a^n \sim \mathrm{id}$ (closed case). Two such quotients are isomorphic if and only if they have the same period n.*

Proof. By Remark 4.7, invertibility forces $k = 0$. The only remaining parameter is n, the period. Cyclic groups $\mathbb{Z}/n_1\mathbb{Z}$ and $\mathbb{Z}/n_2\mathbb{Z}$ are isomorphic iff $n_1 = n_2$. □

Remark 4.13 (The scope of Corollary 4.12)*.* Corollary 4.12 is elementary: it states that cyclic groups are classified by their order. This is a standard result of group theory and contains no new content. The non-trivial classification is Proposition 4.10 for general (non-invertible) systems, where $k \geq 1$ is possible and the (n, k) pair genuinely distinguishes non-isomorphic quotients that have the same number of elements in their kernel cycle but different tail lengths. The examples of pseudo-periodic systems in §7.5 are all in this non-invertible regime.

5 Metric Emergence

Definition 5.1 (Weight Function)**.** A *weight function* is a map $w : R \to [0, \infty)$, extended to configurations by:

$$w(C) = w(r_1) + \cdots + w(r_n) \quad \text{for } C = (r_1, \dots, r_n), \qquad w(\mathrm{id}) = 0.$$

When w takes values in $\mathbb{N}$, it is a *discrete weight function.*

Definition 5.2 (Relational Path Between Objects)**.** A *relational path from* $[A]$ *to* $[B]$ is a configuration $P \in \mathcal{C}$ such that for some representative $a \in [A]$, the configuration $a \circ P$ belongs to $[B]$. The set of all such paths is $\mathrm{Path}([A], [B])$.

Lemma 5.3 (Representative independence)**.** *The set* $\mathrm{Path}([A], [B])$ *and the value* $\inf\{w(P) \mid P \in \mathrm{Path}([A], [B])\}$ *are independent of the choice of representative* $a \in [A]$*.*

Proof. Let $a, a' \in [A]$, so $a \sim a'$. Suppose $P \in \mathcal{C}$ satisfies $a \circ P \in [B]$. Since $a \sim a'$ and $P \sim P$, the congruence property gives $a \circ P \sim a' \circ P$. Hence $a' \circ P \in [B]$, so $P \in \mathrm{Path}([A], [B])$ via a' as well. The set $\mathrm{Path}([A], [B])$ is therefore independent of representative. Independence of the infimum follows immediately, since the infimum is taken over the same set. ☐

Theorem 5.4 (Metric Emergence)**.** *Let* (V, R, s, t) *be an invertible relational system with congruence* $\sim$ *and weight function* $w : R \to [0, \infty)$*. Assume:*

(A1) $w(r) \geq 0$ *for all* $r \in R$*,*

(A2) $w(C_1) = w(C_2)$ *whenever* $C_1 \sim C_2$*,*

(A3) $w(\mathrm{id}) = 0$*,*

(A4) $w(r^{-1}) = w(r)$ *for all* $r \in R$*.*

Define:

$$d([A], [B]) = \inf\bigl\{w(P) \mid P \in \mathrm{Path}([A], [B])\bigr\}.$$

Then d *is a* pseudometric *on* $\mathcal{C}/{\sim}$*. If additionally:*

(A5) $d([A], [B]) = 0 \Rightarrow [A] = [B]$*,*

then d *is a* metric*.*

Proof. **Non-negativity.** Immediate from (A1).

Self-distance. $\mathrm{id} \in \mathrm{Path}([A], [A])$ and $w(\mathrm{id}) = 0$ by (A3), so $d([A], [A]) = 0$.

Symmetry. By invertibility and (A4), reversing $P = (r_1, \dots, r_n)$ yields $P^{-1} = (r_n^{-1}, \dots, r_1^{-1})$ from $[B]$ to $[A]$ with $w(P^{-1}) = w(P)$. Hence $d([A], [B]) = d([B], [A])$.

Triangle inequality. For $P \in \mathrm{Path}([A], [B])$ and $Q \in \mathrm{Path}([B], [C])$, the concatenation $P \circ Q \in \mathrm{Path}([A], [C])$ satisfies $w(P \circ Q) = w(P) + w(Q)$. Taking infima: $d([A], [C]) \leq d([A], [B]) + d([B], [C])$.

Metric. Assumption (A5) gives separation. ☐

Corollary 5.5 (Discrete metric emergence)**.** *If* $w : R \to \mathbb{N}$ *with* $w(r) \geq 1$ *for all* $r \in R$ *and* R *is finite, the infimum in Theorem 5.4 is attained:*

$$d([A], [B]) = \min\bigl\{w(P) \mid P \in \mathrm{Path}([A], [B])\bigr\}.$$

Proof. Any path of k steps has weight at least k. For fixed target weight M, the set of paths with $w(P) \leq M$ is finite (bounded length, finite R). The infimum over a finite non-empty set of non-negative integers is a minimum. ☐

Remark 5.6 (Separation condition). Assumption (A5) is not automatic. If $w(r) = 0$ for some $r \not\sim \mathrm{id}$, then $d([A], [A \circ r]) = 0$ even though $[A] \neq [A \circ r]$. The condition holds whenever $w(r) > 0$ for all $r \in R$.

6 Metric Universality

6.1 Reconstruction of Finite Metric Spaces

Theorem 6.1 (Reconstruction). *Let (X, d) be a finite metric space. Then there exists a relational system (V, R, s, t) with weight function $w : R \to [0, \infty)$ such that the induced metric on $\mathcal{C}/\sim$ is isometric to (X, d).*

Proof. Set $V := X$, $R := \{e_{xy} \mid x, y \in X\}$, and define $s(e_{xy}) := x$, $t(e_{xy}) := y$, and $w(e_{xy}) := d(x, y)$. The congruence $\sim$ is generated by all equalities implied by the metric structure; the identity at x is $\mathrm{id}_x = e_{xx}$ with $w(e_{xx}) = 0$.

The single-edge path (e_{xy}) gives $d_{\mathrm{ind}}(x, y) \leq d(x, y)$. For any path $\gamma = (e_{x_0 x_1}, \ldots, e_{x_{n-1} x_n})$, the triangle inequality for d gives $d(x, y) \leq \sum_i d(x_i, x_{i+1}) = w(\gamma)$. Taking the infimum: $d \leq d_{\mathrm{ind}}$. Hence $d_{\mathrm{ind}} = d$, and the identity map is an isometry. □

Remark 6.2 (Expressibility and canonicity). Theorem 6.1 establishes *expressibility*: every finite metric space encodes as a relational system. It does not establish canonicity: there is no unique encoding. Different relational presentations of the same point set, with different weight functions, correspond to different structural comparisons.

6.2 Minimal Presentations

Definition 6.3 (Redundant generator). A generator $e_{xy} \in R$ is *redundant* if there exists a configuration $\gamma \in \mathrm{Conf}(x, y)$ not using e_{xy} with $w(\gamma) = d(x, y)$.

Theorem 6.4 (Minimal presentation). *Every finite metric space (X, d) admits a generating relational system with no redundant generators, minimal with respect to edge inclusion.*

Proof. Start from the complete-graph construction of Theorem 6.1. Iteratively remove redundant generators; finiteness of X guarantees termination. Removing a redundant generator preserves the induced metric. No remaining generator can be removed without increasing some distance. □

Remark 6.5 (Non-uniqueness of minimal presentations). Minimal presentations are not unique in general. If $d(x, z) = d(x, y) + d(y, z)$ for some intermediate point y, the edge e_{xz} is redundant and may be omitted; otherwise it is not. Non-uniqueness arises precisely where the triangle inequality is tight.

Proposition 6.6 (Minimal presentations and geodesic generators). *A generator e_{xy} is non-redundant in a minimal presentation of (X, d) if and only if the direct step $x \to y$ is a geodesic segment — that is, no path through intermediate points achieves the same weight. Equivalently:*

$$e_{xy} \text{ is non-redundant} \iff d(x, y) < d(x, z) + d(z, y) \text{ for all } z \neq x, y.$$

The non-uniqueness of minimal presentations is governed exactly by the tight triangles of (X, d): for each triple with $d(x, z) = d(x, y) + d(y, z)$, the long edge e_{xz} may be included or omitted, yielding distinct but metrically equivalent minimal presentations.

Proof. A generator e_{xy} is redundant iff there exists an alternative path of equal weight — i.e., $d(x, y) = d(x, z) + d(z, y)$ for some intermediate z. The contrapositive gives non-redundancy iff no such z exists, which is the strict triangle inequality at every intermediate point. Non-uniqueness follows: whenever the triangle inequality is tight at some triple, the long-edge generator is optional. □

6.3 Domain of Validity

Theorem 6.7 (Domain of validity). *The framework captures:*

(i) ***Finite metric spaces (exact)*** *via Theorem 6.1 and the categorical equivalence (Theorem 8.9).*

(ii) ***Compact metric spaces (GH-approximate)*** *via Theorem 9.2: any compact metric space is a Gromov–Hausdorff limit of finite relational systems.*

(iii) ***Locally finite non-compact spaces (pointed GH-approximate)*** *via Theorem 9.4.*

The following lie outside the current scope: non-locally-finite infinite spaces, smooth Riemannian structure (tangent spaces, differential forms), and continuous symmetry groups.

7 Examples

7.1 The Discrete Circle

Let $R = \{a\}$. Fix $n \geq 1$ and define the congruence generated by $a^n \sim \mathrm{id}$.

Verification. This is a congruence: if $k \equiv k' \pmod{n}$ and $\ell \equiv \ell' \pmod{n}$, then $k+\ell \equiv k'+\ell' \pmod{n}$.

Quotient. By Theorem 4.3:

$$\mathcal{C}/\sim\, = \{[a^0], [a^1], \ldots, [a^{n-1}]\} \cong \mathbb{Z}/n\mathbb{Z},$$

with $[a^k] \circ [a^\ell] = [a^{k+\ell \bmod n}]$.

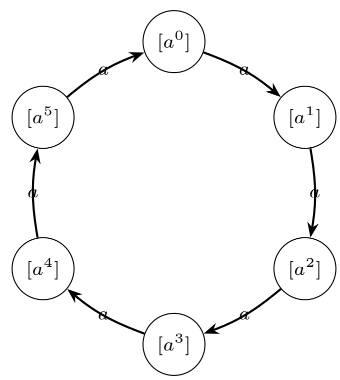

Figure 1: The discrete circle $\mathbb{Z}/6\mathbb{Z}$ as a relational quotient. The law $a^6 \sim \mathrm{id}$ forces closure.

Example 7.1 (Distance on the discrete circle)**.** Assign $w(a) = 1$. The distance between $[a^j]$ and $[a^k]$ is:

$$d\big([a^j], [a^k]\big) = \min\big(|k-j| \bmod n,\ n - (|k-j| \bmod n)\big).$$

The maximum distance is $\lfloor n/2 \rfloor$, reflecting circular topology. No coordinates or embedding are used.

7.2 End-to-End Worked Example: Four-Point Metric Space

Let $X = \{p, q, r, s\}$ with $d(p,q) = d(q,r) = d(r,s) = 1$, $d(p,r) = d(q,s) = 2$, $d(p,s) = 3$: four equally-spaced collinear points.

Starting from the complete-graph presentation (12 edges), the edges $e_{pr}, e_{rp}, e_{qs}, e_{sq}, e_{ps}, e_{sp}$ are all redundant (each has an alternative path of equal weight). The minimal presentation is the path graph $\{e_{pq}, e_{qp}, e_{qr}, e_{rq}, e_{rs}, e_{sr}\}$.

The induced metric recovers d exactly: $d_{\mathrm{ind}}(p,r) = 2$, $d_{\mathrm{ind}}(p,s) = 3$, etc. The triangle inequality is tight at every triple, which is precisely the condition for a unique minimal path presentation.

7.3 The Discrete Torus

Let $R = \{a, b\}$ with the congruence generated by:

$$a \circ b \sim b \circ a, \quad a^n \sim \mathrm{id}, \quad b^m \sim \mathrm{id}.$$

By Theorem 4.3:

$$\mathcal{C}/\sim\, \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}.$$

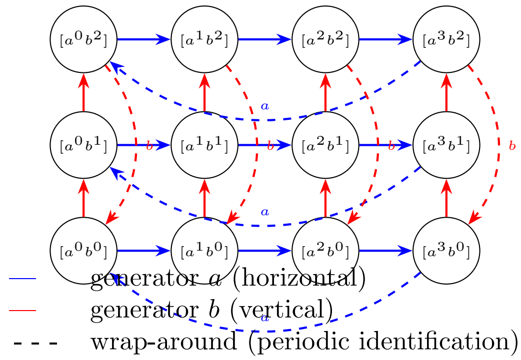

Figure 2: The discrete torus $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Dashed arrows indicate periodic identifications; commutativity $ab \sim ba$ means traversal order is irrelevant.

7.4 Relational Curvature via Noncommutativity

Let $R = \{a, b, a^{-1}, b^{-1}\}$ with $a \circ a^{-1} \sim \mathrm{id}$ and $b \circ b^{-1} \sim \mathrm{id}$. Define the *relational commutator*:

$$[a, b] = a \circ b \circ a^{-1} \circ b^{-1}.$$

If $[a, b] \sim \mathrm{id}$, the loop closes: the structure is locally flat. If $[a, b] \not\sim \mathrm{id}$, traversing the loop $a \circ b \circ a^{-1} \circ b^{-1}$ does not return to the starting class. This is the discrete relational analogue of holonomy: the signature of nonzero curvature.

Curvature is a discrete, finitely computable property: check whether $[a, b] \sim \mathrm{id}$ under the congruence.

Remark 7.2 (Relationship to lattice gauge theory). The relational commutator $[a, b] = a \circ b \circ a^{-1} \circ b^{-1}$ is, in the language of lattice gauge theory, the Wilson loop around an elementary plaquette [4]. When the generators carry values in a Lie group G (the group-valued extension considered in the companion paper), the commutator becomes the face holonomy $g^a \cdot g^b \cdot (g^a)^{-1} \cdot (g^b)^{-1} \in G$, and $[a, b] \sim \mathrm{id}$ is the discrete flatness condition. The identification of the relational commutator with holonomy is not new; it is explicit in Wilson [4]. What the PRG framework adds is the categorical framing: the flatness condition $[a, b] \sim \mathrm{id}$ is a congruence relation on the free category, and quotienting by it is the same operation as restricting to flat discrete connections.

Remark 7.3 (Algebraic vs. metric curvature). Nontrivial commutators detect *failure of local flatness* at the algebraic level: the loop $a \circ b \circ a^{-1} \circ b^{-1}$ does not return to its starting class, which is the discrete analogue of holonomy. However, noncommutativity alone does not imply negative metric curvature: free groups are highly noncommutative yet δ-hyperbolic; nilpotent groups are noncommutative but not hyperbolic. Metric curvature is captured quantitatively by the triangle defect κ of Proposition 7.4. The commutator $[a, b] \not\sim \mathrm{id}$ is a necessary algebraic precondition for curvature; the triangle defect provides the metric measurement. The two notions coincide — a nontrivial commutator produces a nonzero triangle defect — only under additional geometric conditions such as local regularity of the presentation.

Proposition 7.4 (Triangle defect and δ-hyperbolicity)**.** *Let $(\mathcal{O}, d)$ be the metric space induced by a relational system with inverses. Define the* triangle defect*:*

$$\kappa(x, y, z) := d(x, y) + d(y, z) - d(x, z).$$

(i) If $\kappa(x, y, z) \le \delta$ for all $x, y, z \in \mathcal{O}$, then $(\mathcal{O}, d)$ is $O(\delta)$-hyperbolic in the sense of Gromov.

(ii) If $(\mathcal{O}, d)$ is δ'-hyperbolic, then $\kappa(x, y, z) \le \delta'$ for all x, y, z.

In particular, $\kappa \equiv 0$ characterizes tree-like (0-hyperbolic) metric spaces.

Proof. *(i).* Assume $\kappa(x, y, z) \le \delta$ for all triples, i.e., $d(x, z) \ge d(x, y) + d(y, z) - \delta$. Take any geodesic triangle with vertices x, y, z and a point p on side $[x, y]$ with $d(x, p) = a$, $d(p, y) = b$, $a + b = d(x, y)$. Applying the defect bound to (x, p, z) gives $d(p, z) \leq d(x, z) - a + \delta$. Comparing with the path $p \to y \to z$ of length $b + d(y, z)$ and using the defect bound for (x, y, z), a standard thin-triangle argument shows p lies within 2δ of $[x, z] \cup [y, z]$. Hence triangles are 2δ-thin and $(\mathcal{O}, d)$ is $O(\delta)$-hyperbolic. The precise constant relating defect bounds to the thin-triangle constant depends on the chosen definition of hyperbolicity; see [2] for a careful account.

(ii). If $(\mathcal{O}, d)$ is δ'-hyperbolic, apply the four-point condition with base point y: $d(x, z) + d(y, y) \leq d(x, y) + d(y, z) + \delta'$. Since $d(y, y) = 0$, this gives $\kappa(x, y, z) \leq \delta'$. □

7.5 Pseudo-Periodic Examples

Definition 7.5 (Pseudo-Periodic System)**.** A relational system with generator a and congruence $\sim$ is *pseudo-periodic* with period n and shift ϕ if $a^n \sim \phi$ and $\phi \not\sim \mathrm{id}$.

Example 7.6 (Helix)**.** Let $R = \{a, b\}$ with $a^n \sim b$ and $b^k \sim \mathrm{id}$. Each traversal of n steps in a advances one step in b; after k traversals the b-direction closes. The quotient encodes a discrete helix: n steps per turn, k turns before closure.

Remark 7.7 (Connection to semidirect products). If the shift ϕ acts on the cyclic structure by an automorphism φ, the resulting quotient is $\mathbb{Z}/n\mathbb{Z} \rtimes_\varphi \langle\phi\rangle$. The helix, screw motion, and glide reflection are all semidirect products of this form.

7.6 Negative Curvature: The Infinite Regular Tree

Fix $k \geq 2$. Let V be all finite words over $\{1, \ldots, k\}$; for each vertex v and symbol i, include $e_{v,i} : v \to vi$ and $\bar{e}_{v,i} : vi \to v$, with unit weights.

Proposition 7.8. *The induced metric satisfies $d(v, w) = |v| + |w| - 2|v \wedge w|$, where $v \wedge w$ is the longest common prefix.*

Theorem 7.9 (Hyperbolic features)**.** *The k-regular tree satisfies: (i) unique geodesics between any two vertices; (ii) exponential volume growth: k^n vertices at distance n from the root; (iii) 0-thin triangles: each side lies in the union of the other two.*

7.7 Positive Curvature: Triangulated Sphere

Let T be a triangulation of S^2 with vertex set V_T, edge set E_T, and unit weights. The induced metric d_T counts minimum triangulation edges in any path. The diameter is finite.

Example 7.10 (Geodesic non-uniqueness: icosahedral graph)**.** The icosahedral graph has 12 vertices, diameter 3, and $A_5 \times \mathbb{Z}/2\mathbb{Z}$ symmetry. Any vertex has exactly 6 shortest configurations of length 3 to its antipodal vertex: the relational analogue of multiple geodesics between antipodal points on a positively curved surface.

7.8 Flat Geometry: Integer Grid

Let $V_n = \{0, \ldots, n\}^2$ with all 8-connected edges of unit weight.

Proposition 7.11. *The induced metric is the Chebyshev distance: $d_n((i_1, j_1), (i_2, j_2)) = \max\{|i_1 - i_2|, |j_1 - j_2|\}$.*

The two generators a (horizontal) and b (vertical) commute, $a \circ b \sim b \circ a$, confirming $[a, b] \sim \mathrm{id}$: the structure is flat. The torus of Section 7.3 is this grid with periodic boundary conditions.

8 Categorical Equivalence

8.1 Lawvere Metric Spaces

Definition 8.1 (Lawvere metric space [1])**.** A *Lawvere metric space* is a set X with $d : X \times X \to [0, \infty]$ satisfying $d(x, x) = 0$ and $d(x, z) \leq d(x, y) + d(y, z)$. Symmetry is not required in general. A classical metric space is a symmetric Lawvere space with $d(x, y) = 0 \Rightarrow x = y$.

8.2 Relational Systems as Free Enriched Categories

Remark 8.2 (Generators and relations). In categorical language, a relational system (V, R, s, t) with congruence $\sim$ is a *category presented by generators and relations*: R supplies the generating morphisms and $\sim$ supplies the relations imposed on them. The word *relation* now carries its categorical meaning — a law imposed on composites — rather than its set-theoretic meaning. This is why the elements of R are called generators rather than relations throughout the paper.

Definition 8.3 (Free enriched category)**.** Given a weighted relational system (V, R, s, t, w), define a category enriched over $([0,\infty], \geq, +, 0)$ by: objects are states V; hom-value $\mathcal{C}(x,y) := \min_{\gamma\in\mathrm{Conf}(x,y)} w(\gamma)$ (∞ if no path exists). Composition: $\mathcal{C}(x,z) \leq \mathcal{C}(x,y) + \mathcal{C}(y,z)$.

Proposition 8.4. *$(V, R, s, t, w) \mapsto \mathcal{C}$ defines a well-formed $([0,\infty], \geq, +, 0)$-enriched category.*

Theorem 8.5 (Universal property)**.** *The assignment $(V, R, s, t, w) \mapsto \mathcal{C}$ is left adjoint to the forgetful functor from $([0,\infty], \geq, +, 0)$-enriched categories to weighted relational systems.*

Sketch. Given (V, R, s, t, w) and an enriched category $\mathcal{D}$, any weight-preserving map $\phi : R \to \mathcal{D}$ extends uniquely to an enriched functor $\mathcal{C} \to \mathcal{D}$ by sending each configuration to the composite of its images. □

8.3 Metric Realization Functor

Definition 8.6 (Category of relational systems)**. RelSys**: objects are weighted relational systems (V, R, s, t, w); morphisms $F : (V, R, s, t, w) \to (V', R', s', t', w')$ are maps $F_V : V \to V'$ and $F_R : R \to R'$ preserving source, target, and satisfying $w'(F_R(r)) \leq w(r)$ for all $r \in R$.

Theorem 8.7 (Metric realization functor)**.** *There is a functor $\mathcal{M}$: **RelSys** $\to$ **LawMet** defined by $\mathcal{M}(V, R, s, t, w) := (V, d)$ where $d(x,y) = \mathcal{C}(x,y)$.*

Remark 8.8 (Two metric levels and their relationship). The framework carries two distinct metric structures that must be distinguished.

The object metric (Theorem 5.4) is defined on $\mathcal{C}/\sim$: equivalence classes $[C]$ of configurations. For a single-state system ($|V| = 1$), all configurations are endomorphisms, $\mathcal{C}/\sim$ is a quotient monoid or group, and this metric is the word metric of that presentation. The examples of Section 7 (circle, torus, tree) all operate at this level.

The state metric (Definition above) is defined on V: the hom-value $\mathcal{C}(x,y)$ measures the minimum-weight path between states x and y. This is the quantity used in the reconstruction theorem and the categorical equivalence. It is the Lawvere metric structure on the underlying graph.

The bridge: in the canonical complete-graph presentation of Theorem 6.1, where $V = X$ and every pair of states has a direct generator, the state metric $\mathcal{C}(x,y)$ recovers the original metric $d(x,y)$ exactly. For multi-state systems the state metric is the appropriate geometric invariant; the object metric on $\mathcal{C}/\sim$ captures the richer process-equivalence structure within each hom-class. The categorical equivalence (Theorem 8.9) operates at the state metric level; an analogous equivalence at the object metric level would require tracking hom-sets, which is part of the framework's open program.

Theorem 8.9 (Equivalence of categories)**.** *Let* **RelSys**$_\sim$ *be* **RelSys** *with metrically equivalent objects identified. Then:*

$$\mathbf{RelSys}_\sim \ \simeq\ \mathbf{LawMet}_{\mathrm{fin}}.$$

Proof. **Essential surjectivity.** By Theorem 6.1, every finite metric space arises as $\mathcal{M}(V, R, s, t, w)$.

Full faithfulness. *Injectivity.* If two morphisms F_1, F_2 induce the same map on states, they agree. *Surjectivity.* Given 1-Lipschitz $f : (V,d) \to (V',d')$, work in the canonical complete presentations: set $F_V = f$ and $F_R(e_{xy}) = e'_{f(x)f(y)}$. Then $w'(F_R(e_{xy})) = d'(f(x), f(y)) \leq d(x,y) = w(e_{xy})$ by 1-Lipschitz, so F is a morphism inducing f. □

9 Compatibility with Classical Geometry

Definition 9.1 (Metric approximation)**.** A sequence of relational systems (G_n, w_n) with induced metrics (V_n, d_n) and maps $\phi_n : V_n \to X$ *approximates* a compact metric space (X, d) if: (i) $\phi_n(V_n)$ is ε_n-dense in X with $\varepsilon_n \to 0$; (ii) $|d_n(x, y) - d(\phi_n(x), \phi_n(y))| \leq \varepsilon_n$.

Theorem 9.2 (Gromov–Hausdorff bridge [2])**.** *If (G_n, w_n) approximates (X, d), then $d_{\mathrm{GH}}((V_n, d_n), (X, d)) \to 0$.*

Proof. Construct correspondences $\mathcal{R}_n$ from ϕ_n and a near-inverse ψ_n. Both maps satisfy $d(\phi_n(x), z) \leq \varepsilon_n$. The triangle inequality gives distortion $\leq 3\varepsilon_n$, hence $d_{\mathrm{GH}} \leq \frac{3}{2}\varepsilon_n \to 0$. □

Corollary 9.3. *S^1, S^2, and $\mathbb{R}^2$ (locally) arise as Gromov–Hausdorff limits of finite relational systems via $\mathbb{Z}/n\mathbb{Z}$, geodesic triangulations, and integer grids respectively.*

Theorem 9.4 (Pointed GH emergence)**.** *Let (G_n, w_n, v_n) be a sequence of rooted relational systems satisfying: (i) uniform local finiteness (degree $\leq D$); (ii) uniform edge bounds ($a \leq w_n(e) \leq b$); (iii) rooted local convergence (metric balls stabilize up to graph isomorphism for each fixed radius). Then (V_n, d_n, v_n) converges in the pointed Gromov–Hausdorff sense to a pointed metric space (V, d, v).*

Example 9.5 (The k-regular tree as pointed GH limit)**.** Let G_n be the finite ball of radius n in T_k, with root v_0 and unit weights. The sequence satisfies all three hypotheses and converges to (T_k, d, v_0).

10 Multi-Object Configurations

Definition 10.1 (Inter-system relations)**.** Let (G_A, w_A) and (G_B, w_B) be relational systems with disjoint vertex sets V_A, V_B. An *inter-system relational structure* is a weighted directed graph G_{AB} with edges in $V_A \times V_B \cup V_B \times V_A$ and weight function w_{AB}. The *combined relational system* is $(G_A \sqcup G_{AB} \sqcup G_B, w_A \sqcup w_{AB} \sqcup w_B)$.

Definition 10.2 (Combined metric)**.** The *combined metric* on $V_A \cup V_B$ is the induced metric of the combined system: $d(x, y) = \inf_\gamma w(\gamma)$ over all configurations γ in the combined system.

Theorem 10.3 (Space emergence)**.** *Let (G_A, w_A), (G_B, w_B) be finite relational systems and G_{AB} an inter-system structure. Then:*

(i) The combined system induces a finite metric space $(V_A \cup V_B, d)$ fully determined by G_A, G_B, and G_{AB}.

(ii) Different choices of G_{AB} over the same G_A, G_B produce genuinely different combined geometries.

Proof. (i) The combined system is a finite relational system; Theorem 5.4 applies. (ii) Different E_{AB} and w_{AB} produce different path sets, hence different infima. □

Special cases: $E_{AB} = \emptyset$ gives metrically disconnected union; a single bridge gives dumbbell geometry; zero-weight connections give metric gluing (relational pushout); uniform bipartite connections give product-like structure; position-varying weights give discrete warped products.

11 Relationship to Existing Work

The mathematical structures in this paper belong to a well-developed tradition. We state explicitly what is reformulated and what is new.

What is reformulated. A relational system (V, R, s, t) with congruence $\sim$ is a category presented by generators and relations in the standard sense [8, 10]. The quotient construction (Theorem 4.3) is the universal property of such a presentation. The metric emergence theorem (Theorem 5.4) is, in the single-state case, the word metric on a group presentation; in the multi-state case it is the path metric of the Cayley graph of the quotient groupoid. Both are classical objects in geometric group theory [11]. The categorical equivalence (Theorem 8.9) is a consequence of Lawvere's identification of metric spaces with categories enriched over $([0, \infty], \geq, +, 0)$ [1], combined with the reconstruction theorem. The Gromov–Hausdorff bridge results (Section 9) are standard applications of the GH framework [2].

What is new. The formulation of the (n, k) classification in relational-process language is a restatement of the classical theory of cyclic monoids [3] within the four-layer framework (generators, congruence, quotient, metric). As a mathematical result, it is an instance of Clifford and Preston's classification of cyclic semigroups by index and period [3]. The *specific contribution* is making this classical result visible inside a framework that separates generators from congruences from quotients, which allows its scope to be stated precisely: the classification applies to general (non-invertible) single-generator systems; for invertible systems the result collapses to the elementary fact that cyclic groups are classified by order (Corollary 4.12). The pointed GH emergence theorem (Theorem 9.4) with its three explicit conditions is a concrete relational formulation not present in the metric geometry literature. The domain of validity theorem (Theorem 6.7), which organizes the boundary between exactly representable, GH-approximable, and out-of-scope spaces, is new as a unified statement.

The identification of the relational commutator with discrete holonomy (Section 7.4) follows Wilson [4] and is not new.

The honest summary. The machinery is classical. The primary contribution is a presentation-level semantic framework: geometric structure is defined at the level of generators and congruence laws, prior to quotienting, and metric properties arise as invariants of congruence classes of processes. The framework isolates four layers — generators (syntax), congruence (laws), quotient (objects), metric (emergent distance) — that are usually conflated in geometric group theory expositions. The framework sits within the tradition of geometric group theory, graph geometry, and enriched category theory as one further development, not as a replacement for those foundations. The induced metric coincides with the path metric on the Cayley graph of the presented groupoid.

An open question answered in Section 12. The four-layer separation makes explicitly askable a question that classical treatments obscure, since they typically begin from the quotient: *what homological structure does a specific congruence law produce in the nerve of the free category?* Section 12 develops this question into a concrete construction — the depth filtration of the bar complex — and proves three theorems about its persistence diagram. The result is a computable invariant of any finite group presentation that records when each law first reshapes the topology of the classifying space.

12 Presentation-Level Persistent Homology

12.1 Motivation

The preceding sections define the quotient group $G = F(R)/\sim$ and compute its homology $H_*(G; \mathbb{Z}) = H_*(BG; \mathbb{Z})$ via the nerve of the quotient category. This is a single object: the topology of BG as a whole. It does not record when each congruence law in $\mathcal{L}$ first affects the topology, nor which laws are responsible for which homology classes.

The construction of this section addresses that gap. By treating word length as a filtration parameter, we obtain a nested sequence of simplicial complexes whose persistent homology tracks the birth and death of homology classes as the nerve grows. Comparing the filtration of the free group nerve with the filtration of the quotient nerve isolates, for each congruence law, the depth at which it first produces a topological effect.

This is not classical group homology: it is a finer invariant that sees the *process* by which the topology of BG assembles from the topology of $BF(R)$ under the successive imposition of laws.

12.2 The Bar Complex and its Depth Filtration

We work throughout with a finite group presentation $\langle R \mid \mathcal{L} \rangle$, where R is a finite set of generators (with inverses named separately), $\mathcal{L}$ is a finite set of relational laws $\{l_i \sim r_i\}$ generating the congruence $\sim$, and $G = F(R)/\sim$ is the presented group.

Definition 12.1 (Nerve of a single-object category)**.** Let M be a monoid (a category with a single object $*$). The *nerve* $N(M)$ is the simplicial set whose n-simplices are n-tuples $(g_1,\ldots,g_n)\in M^n$, with face maps:

$$d_i(g_1,\ldots,g_n)=\begin{cases}(g_2,\ldots,g_n) & i=0,\\ (g_1,\ldots,g_ig_{i+1},\ldots,g_n) & 0<i<n,\\ (g_1,\ldots,g_{n-1}) & i=n,\end{cases}$$

and degeneracy maps inserting the identity. The geometric realization $|N(M)|$ is the classifying space BM. For a group G, $H_n(|N(G)|;\mathbb{Z})=H_n(G;\mathbb{Z})$ (group homology with trivial coefficients) by the standard identification [9].

Remark 12.2 (Bar complex). The cellular chain complex of $|N(G)|$ is the *normalized bar complex* $\bar{B}_*(G)$: the chain group $\bar{B}_n$ is the free abelian group on G^n (modulo degeneracies), and the boundary operator is $\partial_n=\sum_{i=0}^{n}(-1)^i d_i$. The identity $\partial_{n-1}\circ\partial_n=0$ follows from the simplicial identity $d_i\circ d_j=d_{j-1}\circ d_i$ for $i<j$, which is verified directly from the face map formula. Computing $H_n(G;\mathbb{Z})$ via Smith normal form of the boundary matrices is the method implemented in Section 12.6.

Definition 12.3 (Depth filtration)**.** Let $d\geq 1$ be a positive integer (the *nerve depth*).

(i) The *free filtration at depth* d is the simplicial subcomplex $\mathcal{F}_d\subseteq N(F(R))$ whose n-simplices are n-tuples $(w_1,\ldots,w_n)$ of freely-reduced words such that each w_i has length at most d and all pairwise and iterated compositions w_iw_{i+1}, $w_iw_{i+1}w_{i+2}$, etc., are also freely reduced with length at most d. The underlying vertex set is $F(R)_{\leq d}$: all freely-reduced words of length $\leq d$.

(ii) The *quotient filtration at depth* d is the analogous subcomplex $\mathcal{Q}_d\subseteq N(G)$ with vertices the set $G_{\leq d}$ of elements of G whose canonical normal form (under the term-rewriting system associated to $\mathcal{L}$) has length at most d.

By construction, $\mathcal{F}_d\hookrightarrow\mathcal{F}_{d+1}$ and $\mathcal{Q}_d\hookrightarrow\mathcal{Q}_{d+1}$ are simplicial inclusions for all d. The congruence $\sim$ induces a simplicial map $\varphi_d:\mathcal{F}_d\to\mathcal{Q}_d$ (sending each word to its normal form) commuting with the inclusions.

Definition 12.4 (Persistence module and persistence diagram)**.** A *persistence module* over $\mathbb{Z}_{\geq 1}$ is a sequence of abelian groups $\{H_n(\mathcal{Q}_d)\}_{d\geq 1}$ together with the homomorphisms induced by the inclusions $\mathcal{Q}_d\hookrightarrow\mathcal{Q}_{d+1}$.

A homology class $\alpha\in H_n(\mathcal{Q}_d)$ is *born* at depth d if it is not in the image of $H_n(\mathcal{Q}_{d-1})\to H_n(\mathcal{Q}_d)$. It *dies* at depth $e>d$ if its image in $H_n(\mathcal{Q}_e)$ is zero but its image in $H_n(\mathcal{Q}_{e-1})$ is nonzero. If its image is nonzero for all e, it has *infinite persistence* and is recorded as the pair (d,∞).

The *persistence diagram* $\mathrm{PD}(\mathcal{Q},n)$ is the multiset of pairs $\{(\mathrm{birth}(\alpha),\mathrm{death}(\alpha))\}$ ranging over all H_n-classes. We write $\mathrm{PD}(\mathcal{F},n)$ for the analogous diagram of the free filtration.

12.3 Three Structural Theorems

Theorem 12.5 (Persistence of the free group filtration)**.** *Let $F(R)$ be the free group on $|R|=r$ generators. The persistence diagram of the free filtration satisfies:*

(i) $\mathrm{PD}(\mathcal{F},0)=\{(1,\infty)\}$*: one class in H_0 born at depth 1, with infinite persistence (connectedness of $BF(R)$).*

(ii) $\mathrm{PD}(\mathcal{F},1)=\{(1,\infty)^r\}$*: exactly r classes in H_1, each born at depth 1 with infinite persistence.*

(iii) $\mathrm{PD}(\mathcal{F},n)=\emptyset$ *for all $n\geq 2$.*

Proof. (i). The complex $\mathcal{F}_d$ is connected for every $d\geq 1$ because the identity element $\mathrm{id}\in F(R)_{\leq d}$ serves as a basepoint and every element can be connected to id via a path through intermediate words. So $H_0(\mathcal{F}_d)\cong\mathbb{Z}$ for all d, and the unique class is born at depth 1 and persists indefinitely.

(ii). At depth $d=1$, the complex $\mathcal{F}_1$ has one vertex for each element of $F(R)_{\leq 1}=\{\mathrm{id},a_1,a_1^{-1},\ldots,a_r,a_r^{-1}\}$ and edges for composable pairs whose composition has length ≤ 1. Each generator a_i gives a free loop (a

class in H_1 based at id), and these r classes are independent. They persist at all depths: for any d, the inclusion $\mathcal{F}_1 \hookrightarrow \mathcal{F}_d$ induces an injection on H_1 because no free reduction ever trivializes a generator.

(iii). Free groups have cohomological dimension 1 [13]: every free group admits a projective resolution of length 1 (the augmented bar complex truncates after the first term), so $H_n(F(R);\mathbb{Z}) = 0$ for $n \geq 2$. Since $H_n(\mathcal{F}_d) \subseteq H_n(F(R))$ for all d, no H_n-classes appear for $n \geq 2$. ☐

Theorem 12.6 (Commutativity creates H_2 at depth 2)**.** *Let $G = \langle a, b \mid a \cdot b \sim b \cdot a\rangle \cong \mathbb{Z}^2$. Then:*

(i) The free filtration $\{\mathcal{F}_d\}$ satisfies $H_2(\mathcal{F}_d) = 0$ for all d.

(ii) The quotient filtration $\{\mathcal{Q}_d\}$ satisfies $H_2(\mathcal{Q}_1) = 0$ and $H_2(\mathcal{Q}_d) \cong \mathbb{Z}$ for all $d \geq 2$.

(iii) The H_2 class born at depth 2 has infinite persistence and arises from the commutativity law.

Proof. (i) is part (iii) of Theorem 12.5.

(ii). At depth 1, $G_{\leq 1} = \{\mathrm{id}, a, a^{-1}, b, b^{-1}\}$ and the complex $\mathcal{Q}_1$ is a graph; in particular, $H_2(\mathcal{Q}_1) = 0$ since simplicial complexes of dimension at most 1 have trivial H_2.

At depth 2, the quotient contains elements such as $[ab] = [ba]$ (identified by the commutativity law). The 2-simplex $([a], [b])$ has boundary $\partial([a], [b]) = [b] - [ab] + [a]$. The identifications $[ab] = [ba]$ create a cycle in the boundary matrix that did not exist in the free filtration, and Smith normal form gives $H_2(\mathcal{Q}_2) \cong \mathbb{Z}$. This is consistent with $H_2(\mathbb{Z}^2;\mathbb{Z}) = H_2(T^2;\mathbb{Z}) \cong \mathbb{Z}$ (the fundamental class of the torus).

(iii). The class born at depth 2 maps under the inclusion $\mathcal{Q}_2 \hookrightarrow \mathcal{Q}_d$ to a nonzero class for all d (it is detected by the standard generator of $H_2(\mathbb{Z}^2)$, which is a global invariant of G). Its birth depth 2 equals the length of the commutativity law $b \cdot a \sim a \cdot b$ (maximum of the two sides), confirming that it is created by the law at the first depth where the law's defining simplex appears. ☐

Theorem 12.7 (Torsion laws modify H_1 persistence)**.** *Let $G = \langle a \mid a^n \sim \mathrm{id}\rangle \cong \mathbb{Z}/n\mathbb{Z}$. Then:*

(i) In the free filtration, the generator a contributes a class in $H_1(\mathcal{F}_d)$ for all $d \geq 1$ with infinite persistence.

(ii) In the quotient filtration, $H_1(\mathcal{Q}_d) \cong 0$ for $1 \leq d < n$, and $H_1(\mathcal{Q}_d) \cong \mathbb{Z}/n\mathbb{Z}$ for all $d \geq n$.

(iii) The torsion class in $H_1(\mathcal{Q}_n)$ is born at depth n (the length of the torsion law) and has infinite persistence.

Proof. (i) follows from Theorem 12.5(ii) applied to the free group on one generator.

(ii). At depth $d < n$, the elements $\mathrm{id}, [a], [a^2], \ldots, [a^{n-1}]$ are n distinct elements of G, and the complex $\mathcal{Q}_d$ for $d < n$ does not yet contain the simplex that witnesses the relation $a^n \sim \mathrm{id}$ (since that simplex is an n-tuple involving a composed n times, which only enters at depth n). For $d < n$, the structure of $\mathcal{Q}_d$ coincides with $\mathcal{F}_d$ restricted to the visible elements. The torsion law first creates the identifying simplex at depth n: the 1-chain $[a^n] = [\mathrm{id}]$ induces a boundary that closes the generator's loop. Smith normal form then gives $H_1(\mathcal{Q}_n) \cong \mathbb{Z}/n\mathbb{Z}$, consistently with $H_1(\mathbb{Z}/n\mathbb{Z};\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$.

(iii). The torsion class is born at depth n because for $d < n$ the law $a^n \sim \mathrm{id}$ is not yet visible in the truncated nerve (the word a^n has length n, so it first enters $G_{\leq d}$ at $d = n$). The class persists for all $d \geq n$ because it reflects the global torsion of G. ☐

Remark 12.8 (Birth depth and law length)*.* Theorems 12.6 and 12.7 both confirm the following pattern: a law $l \sim r$ of *law length* $\ell := \max(|l|, |r|)$ can only produce a topological effect at depth $d \geq \ell$, since the simplices whose boundaries witness the law first appear at depth ℓ. This gives a lower bound on the birth depth of any homology class created by the law. The bound is sharp in both cases above: commutativity (law length 2) creates H_2 at depth 2; the torsion law $a^n \sim \mathrm{id}$ (law length n) creates torsion at depth n.

12.4 The Presentation Persistence Diagram

Definition 12.9 (Presentation persistence diagram)**.** Let $\langle R \mid \mathcal{L}\rangle$ be a finite group presentation with quotient group G. The *presentation persistence diagram in degree n* is the formal difference:

$$\Delta\mathrm{PD}_n(\langle R \mid \mathcal{L}\rangle) \;:=\; \mathrm{PD}(\mathcal{Q}, n) \;\setminus\; \mathrm{PD}(\mathcal{F}, n),$$

where the difference is taken in the sense of multisets: classes in $\mathrm{PD}(\mathcal{F}, n)$ that persist unchanged into $\mathrm{PD}(\mathcal{Q}, n)$ are removed, and only the classes that change (die, are born, or shift in persistence) are retained.

Remark 12.10 (Presentation invariant, not group invariant). The notation $\Delta\mathrm{PD}_n(\langle R \mid \mathcal{L}\rangle)$ is intentional: the argument is the *presentation*, not the group G. This is not a notational convenience; it is a mathematical necessity.

The depth filtration of Definition 12.3 depends on two choices that are not intrinsic to G:

(i) The generating set R and its ordering (which determines the shortlex term order orienting the rewriting rules). Different orderings of the same R produce different normal forms and hence different element depths $G_{\leq d}$.

(ii) The law lengths $\max(|l|, |r|)$ for $l \sim r \in \mathcal{L}$. By Remark 12.8, every homology class created by a law $l \sim r$ has birth depth $\geq \max(|l|, |r|)$. A presentation with longer laws produces larger birth depths, even if it presents the same group.

Concrete example. Consider two presentations of $\mathbb{Z}^2$:

$$P_1 = \langle a, b \mid ba = ab \rangle \qquad \text{(law length 2)}$$
$$P_2 = \langle a, b \mid b^2a^2 = a^2b^2,\ bab^{-1} = a \rangle \qquad \text{(law lengths 4 and 3)}$$

Both present $\mathbb{Z}^2$, so $H_2(\mathbb{Z}^2; \mathbb{Z}) \cong \mathbb{Z}$ in both cases. But Theorem 12.6 applies to P_1 and gives $\mathrm{birth}_{H_2} = 2$; under P_2 the same class cannot be detected before depth 3 (the length of the shorter law). The group homology is the same; the presentation persistence diagrams differ.

What is a group invariant. The *stable* output of the filtration is a group invariant: $H_n(G; \mathbb{Z})$ is recovered from $H_n(\mathcal{Q}_d)$ for all sufficiently large d (for finite groups, once $G_{\leq d} = G$; for infinite groups, in the Gromov–Hausdorff limit). The stable value does not depend on the presentation. What the presentation determines is the *depth at which* each class first appears — the birth function — not whether the class exists.

Consequence for the open question (Q1) of Section 12.7. Any stability theorem for $\Delta\mathrm{PD}_n$ (under perturbation of $\mathcal{L}$) must be stated in terms of a metric on *presentations*, not on *groups*. The natural metric is the law-length metric: two presentations P, P' of G are at distance $\delta(P, P') = \max_{l \sim r \in \mathcal{L} \triangle \mathcal{L}'} \max(|l|, |r|)$, and the expected stability bound is $d_{\mathrm{bottleneck}}(\Delta\mathrm{PD}_n(P), \Delta\mathrm{PD}_n(P')) \leq \delta(P, P')$.

Remark 12.11 (Interpretation). The presentation persistence diagram records precisely what the congruence laws $\mathcal{L}$ contribute to the topology:

- A class in $\mathrm{PD}(\mathcal{F}, n)$ that is absent from $\mathrm{PD}(\mathcal{Q}, n)$ was *killed* by the laws (its free loop was trivialised or made torsion).
- A class in $\mathrm{PD}(\mathcal{Q}, n)$ that is absent from $\mathrm{PD}(\mathcal{F}, n)$ was *created* by the laws (a new cycle appeared that did not exist in the free group nerve).
- The birth depth of a created class is the depth at which the responsible law first enters the filtration, which by Remark 12.8 equals the law's length.

In this sense, $\Delta\mathrm{PD}_n$ decomposes the topology of BG into the contribution of each law, ordered by law length. No existing tool in computational algebraic topology computes this comparison as a first-class operation; classical software computes $H_*(G)$ from the quotient alone.

12.5 Examples

Example 12.12 (Circle: $G = \mathbb{Z}$)**.** Presentation: $\langle a \mid \varnothing \rangle$ (no laws). The quotient is the free group itself: $Q_d = F_d$ for all d. Both persistence diagrams coincide, and $\Delta\mathrm{PD}_n = \emptyset$ for all n. The computation stabilizes at depth 1: $H_1(\mathcal{Q}_1) \cong \mathbb{Z}$, $H_n = 0$ for $n \geq 2$.

Example 12.13 (Torus: $G = \mathbb{Z}^2$)**.** Presentation: $\langle a, b \mid b \cdot a \sim a \cdot b \rangle$.

Degree	$\mathrm{PD}(\mathcal{F}, n)$	$\mathrm{PD}(\mathcal{Q}, n)$	$\Delta\mathrm{PD}_n$
H_0	$(1, \infty)$	$(1, \infty)$	$\emptyset$
H_1	$(1, \infty)^2$	$(1, \infty)^2$	$\emptyset$
H_2	$\emptyset$	$(2, \infty)$	$\{(2, \infty)\}$

The commutativity law creates one H_2 class at depth 2 (law length 2) with infinite persistence. The two generator loops in H_1 are unchanged. The H_2 class $(2, \infty)$ is the fundamental class of the torus.

Example 12.14 (Cyclic group: $G = \mathbb{Z}/3\mathbb{Z}$)**.** Presentation: $\langle a \mid a^3 \sim \mathrm{id}\rangle$.

Degree	$\mathrm{PD}(\mathcal{F}, n)$	$\mathrm{PD}(\mathcal{Q}, n)$	$\Delta\mathrm{PD}_n$
H_0	$(1, \infty)$	$(1, \infty)$	$\emptyset$
H_1	$(1, \infty)$	$(3, \infty)_{\mathbb{Z}/3}$	$\{(1, \infty) \to (3, \infty)_{\mathbb{Z}/3}\}$

The notation $(3, \infty)_{\mathbb{Z}/3}$ indicates a class with $\mathbb{Z}/3\mathbb{Z}$ coefficient born at depth 3. The free H_1 class (born at depth 1, integer coefficient) is transformed into a torsion class at depth 3 (the length of the law $a^3 \sim \mathrm{id}$).

12.6 Implementation

The construction of Section 12 is implemented as a browser-based tool that computes both filtrations from a group presentation entered as generators and relational laws.

Architecture. The implementation has four layers: (1) **Algebra**: a term-rewriting system with shortlex ordering computes canonical normal forms for elements of G. Local confluence is verified by checking all critical pairs (suffix-prefix overlaps and containment overlaps of rewrite rules) for joinability; non-confluence is flagged before nerve construction, as a non-confluent TRS can produce different normal forms for congruent words, corrupting the nerve. (2) **Nerve**: both $\mathcal{F}_d$ and $\mathcal{Q}_d$ are built by enumerating all freely-reduced (resp. normally-reduced) words of length $\leq d$, forming all composable tuples, and recording the resulting simplices. (3) **Linear algebra**: Smith normal form of integer boundary matrices computes $H_n(\mathcal{F}_d)$ and $H_n(\mathcal{Q}_d)$ for each d. (4) **Stability**: depth is treated as a filtration parameter. At each depth d, the homology groups are compared with depth $d-1$; if they agree, the computation has stabilized (homology stability, not element-count stability, is the correct criterion, since infinite groups always have elements beyond any finite depth).

The comparison output. The tool displays both columns of the comparison simultaneously: the free filtration on the left and the quotient filtration on the right, with signed deltas per dimension. This is the presentation persistence diagram in computable form. No existing software package (GAP, SageMath, or Magma) produces this comparison as a single operation from a group presentation; in those systems the free group homology and the quotient homology must be computed and compared manually.

Correctness. The chain complex condition $\partial^2 = 0$ is verified automatically at each depth, serving as an internal consistency check. The Smith normal form algorithm is exact (integer arithmetic), so no floating-point error enters the homology computation.

12.7 Open Questions

(Q1) **Stability.** Does the presentation persistence diagram satisfy a stability theorem in the sense of topological data analysis? Since $\Delta\mathrm{PD}_n$ is a presentation invariant rather than a group invariant (Remark 12.10), stability must be stated in terms of a metric on presentations. The natural candidate is the law-length metric $\delta(P, P') = \max_{l \sim r \in \mathcal{L} \triangle \mathcal{L}'} \max(|l|, |r|)$, and the conjectured stability bound is $d_{\mathrm{bottleneck}}(\Delta\mathrm{PD}_n(P), \Delta\mathrm{PD}_n(P')) \leq \delta(P, P')$. The standard stability theorem for persistent homology of filtered simplicial complexes applies [14] when the interleaving distance between the two filtrations is controlled; quantifying that interleaving in terms of law-length changes is open.

(Q2) **Fox calculus connection.** The Fox calculus provides a chain-level computation of $H_1(G)$ and $H_2(G)$ directly from the presentation matrix [5]. Is the presentation persistence diagram expressible in terms of the Fox derivatives of the relators? A positive answer would connect the depth filtration to existing algebraic invariants of presentations and give a purely algebraic description of birth depths.

(Q3) **Non-invertible extension.** The construction extends to monoid presentations $\langle R \mid \mathcal{L}\rangle$ without inverses, where the nerve is the classifying space of the monoid. The free monoid R^* has a contractible classifying space ($H_n = 0$ for $n \geq 1$), so all nontrivial homology in the quotient monoid filtration is created by the laws. In this case the presentation persistence diagram captures the full homology of BM. The depth filtration for finite cyclic monoids $M(n, k)$ (classified in Section 4.2) is a concrete first family of examples.

(Q4) **Multi-law attribution.** When $|\mathcal{L}| \geq 2$, multiple laws interact: a homology class may be created by the joint action of two laws neither of which alone produces it. Define a partial order on $\mathcal{L}$ by "law l_i is responsible for class α" and investigate when this order is a total order, when laws act independently, and when they exhibit interaction. This is the homological analogue of the problem of identities among relations [5].

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